

Supersnowflakes: Stepwise Self-Assembly and Dynamic Exchange of Rhombus Star-Shaped Supramolecules

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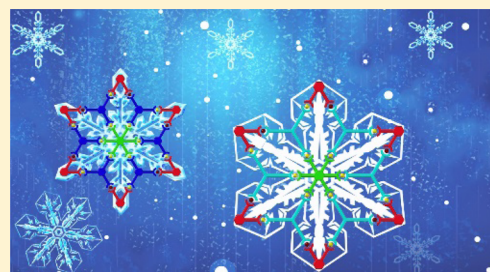
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S Supporting Information

ABSTRACT: With the goal of increasing the complexity of metallo-supramolecules, two rhombus star-shaped supramolecular architectures, namely, supersnowflakes, were designed and assembled using multiple 2,2':6',2''-terpyridine (tpy) ligands in a stepwise manner. In the design of multicomponent self-assembly, ditopic and tritopic ligands were bridged through Ru(II) with strong coordination to form metal–organic ligands for the subsequent self-assembly with a hexatopic ligand and Zn(II). The combination of Ru(II)–organic ligands with high stability and Zn(II) ions with weak coordination played a key role in the self-assembly of giant heteroleptic supersnowflakes, which encompassed three types of tpy-based organic ligands and two metal ions. With such a stepwise strategy, the self-sorting of individual building blocks was prevented from forming the undesired assemblies, e.g., small macrocycles and coordination polymers. Furthermore, the intra- and intermolecular dynamic exchange study of two supersnowflakes by NMR and mass spectrometry revealed the remarkable stability of these giant supramolecular complexes.



INTRODUCTION

Self-assembly as a powerful bottom-up approach has been extensively used by nature to create complex biological systems. Synthetic supramolecular chemistry aims to use the principles of biological self-assembly to construct artificial nanostructures using molecular building blocks with desired functions and properties.¹ Among this diverse research field, coordination-driven self-assembly,^{2–4} a burgeoning section of supramolecular chemistry, is flourishing for the construction of a vast variety of structures, such as helicates,⁵ grids,⁶ polygons,^{7–9} polyhedra,^{10–12} rotaxanes,¹³ catenanes,^{14–17} knots,¹⁸ and links.¹⁹ On the basis of their precisely controlled shapes and sizes, these spectacular metallo-supramolecules have found a wide array of applications, including catalysis,^{16a,20} sensing,²¹ drug delivery and release,²² gas storage,²³ smart materials,^{24,25} and stabilization of reactive substances.²⁶ Most of these achievements, however, were centered on highly symmetric metallo-supramolecules assembled by a single type of organic ligand and

metal ion. The corresponding abiological assemblies still suffered from a lack of complexity and thus were unable to reach the high degrees of functionality found in natural systems.

To advance supramolecular chemistry into a new level of sophistication, efforts have been made to create more complex metallo-supramolecular architectures and ultimately to achieve new chemistry and functionality. To date, the following strategies have been used to increase structural complexity and diversity issues: (i) rational design of ligands with self-sorting ability for multicomponent self-assembly;²⁷ (ii) synthesis of multiarmed building blocks for self-assembly of multilayered structures;²⁸ (iii) stepwise self-assembly of heteroleptic architectures using metal–organic ligands;²⁹ (iv) employing both coordination and dynamic covalent chemistry for subcomponent self-assembly;^{12b,30} (v) introducing other

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types of weak interactions for orthogonal self-assembly in addition to coordination.^{24a,25b,31}

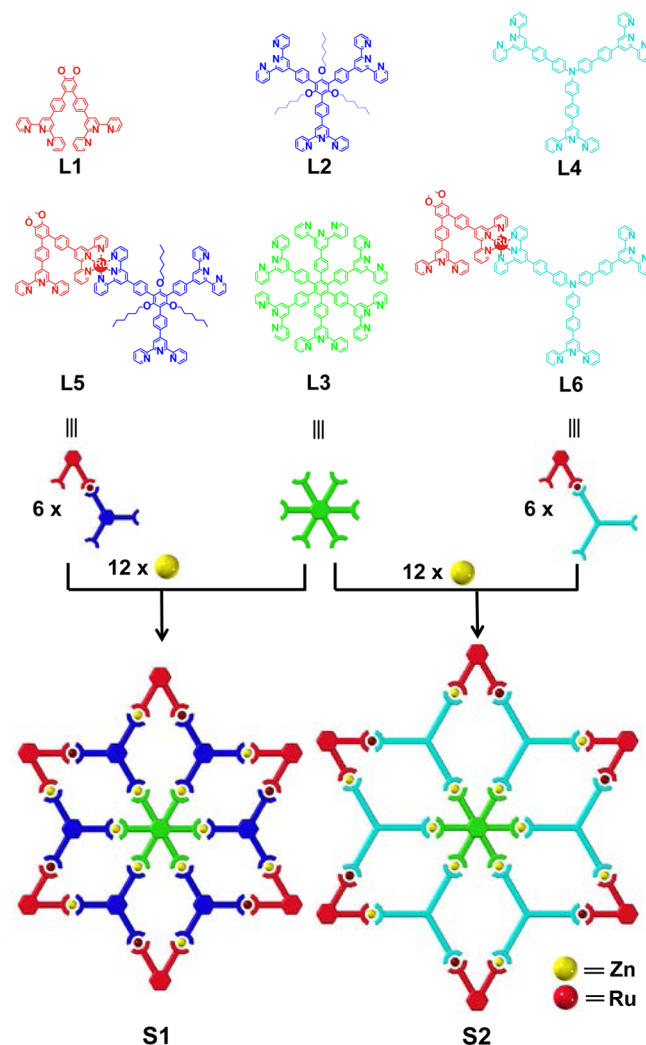
2,2':6',2''-Terpyridine (tpy) ligands are widely used as building blocks in supramolecular and macromolecular chemistry.^{2a,32} An extremely interesting aspect of this tridentate ligand is the different binding strengths of tpy and transition metal ions which order as Ru(II) > Fe(II) > Ni(II) > Zn(II) > Cd(II) in the complex of $[M(tpy)_2]^{2+}$.³³ This main characteristic allows us to harness strong and weak binding metal ions, e.g., Ru(II) with Fe(II), Zn(II), or Cd(II), in a stepwise manner, in which a stable Ru(II)–organic ligand was assembled or synthesized for the subsequent self-assembly with metal ions with low binding strength and high reversibility to construct heteroleptic metallo-supramolecules with increasing complexity.³⁴ With this stepwise approach, most of the previous studies, however, were limited to the combination of two types of ligands and metal ions.^{9a,29b–d,35} Challenges still remain by using three or more types of tpy-based organic ligands in the journey toward multicomponent self-assembly.

Beyond those boundaries, we describe herein a stepwise self-assembly of giant rigid two-dimensional (2D) rhombus star-shaped supramolecules, namely, supersnowflakes, which encompass three types of tpy ligands and two metal ions, i.e., Ru(II) and Zn(II), in each individual structure. It is worthy to note that the pioneering snowflake-like structures by Stang³⁶ and Newkome³⁷ were based on the exofunctionalization of supramolecular hexagons using dendrimers, while our shape-persistent snowflakes were constructed with all rigid backbones. Considering that tpy-based building blocks have been introduced into infinite 2D metal–organic monolayer sheets,³⁸ our discrete and shape-persistent 2D supramolecular architectures may serve as a model system to investigate the self-assembly behavior and physical properties to generate defect-free 2D materials for molecular electronics, energy storage, imaging, and sensing.

RESULTS AND DISCUSSION

Synthesis, Self-Assembly, and Characterization of Supersnowflake 1 (S1). On the basis of structural analysis, three tpy-based organic ligands, L1, L2, and L3,³⁹ may assemble with metal ions with a ratio of 6:6:1:18 to generate a snowflake-like metallo-supramolecule (Scheme 1; Scheme S1, Supporting Information). All three ligands were prepared through conventional Suzuki cross-coupling reactions. Direct self-assembly was conducted by successively adding a methanol solution of $Zn(NO_3)_2$ (18 equiv) into a chloroform solution of L1/L2/L3 (6:6:1) with shaking. Although the concentration of the ligands and Zn(II) solution was low (1 mg/mL), a precipitate was immediately formed after the mixing. The predominant assemblies in the supernatant were identified as small triangles by electrospray ionization mass spectrometry (ESI-MS) (Figure S1, Supporting Information; the counterions were transformed to PF_6^-) due to the strong self-sorting ability of L1. We speculated that the precipitate was the 2D coordination polymers assembled by ligands L2 and L3 with Zn(II).³⁸ To block the self-sorting individual ligands, ruthenium chemistry was employed to prepare L5 by bridging L1 and L2 (Scheme S3, Supporting Information) in the first step. Briefly, L5 was synthesized by Suzuki coupling reaction on a precursor, 7, owing to the excellent chemical stability of tpy–Ru(II) complexes during conventional coupling reaction and column separation. Note that L5 was not able to be obtained by direct mixing of L1, L2, and Ru(II) because of self-sorting and

Scheme 1. Self-Assembly of Snowflakes S1 and S2



separation challenges. In the second step, Ru(II)–organic ligand L5 was assembled with a hexatopic ligand, L3, and $Zn(NO_3)_2$ in an exact ratio of 6:1:12 in $CHCl_3/MeOH$ at 50 °C for 5 h. The red precipitate of S1 was obtained by adding an excess amount of NH_4PF_6 into the assembly mixture and purified by repeated washing with deionized (DI) water.

Figure 1 shows the 1H NMR spectra of ligands L3 and L5 and their corresponding complex S1. Sharp and well-split signals can be seen in the spectra of L3 and L5, which are assigned to one and four sets of tpy–phenyl units, respectively, according to the characteristic resonance peaks of $tpyH^{3',5'}$. After complexation, S1 shows a slightly broad NMR pattern in the aromatic region with six types of tpy species (A–F), due to the overlap of the resonance signals ascribed to the protons in each ligand. All the $3',5'$ protons on the metal-free tpy units of L3 and L5, i.e., $A(E)^{3',5'}$, $D^{3',5'}$, and $F^{3',5'}$, were significantly shifted downfield (~ 0.3 ppm) after coordination with the Zn(II) ion, owing to the electron deficiency of these protons brought about by the metal ions.⁴⁰ In contrast, all the $6,6''$ protons of the above-mentioned tpy groups were dramatically shifted upfield (~ 0.8 ppm), due to the electron shielding effect.⁴⁰ The methoxyl groups of L5 showed distinguished NMR peaks because of their various distances from the metal ion. While in the complex, the corresponding signal was a singlet and was relatively sharp, suggesting the combination of

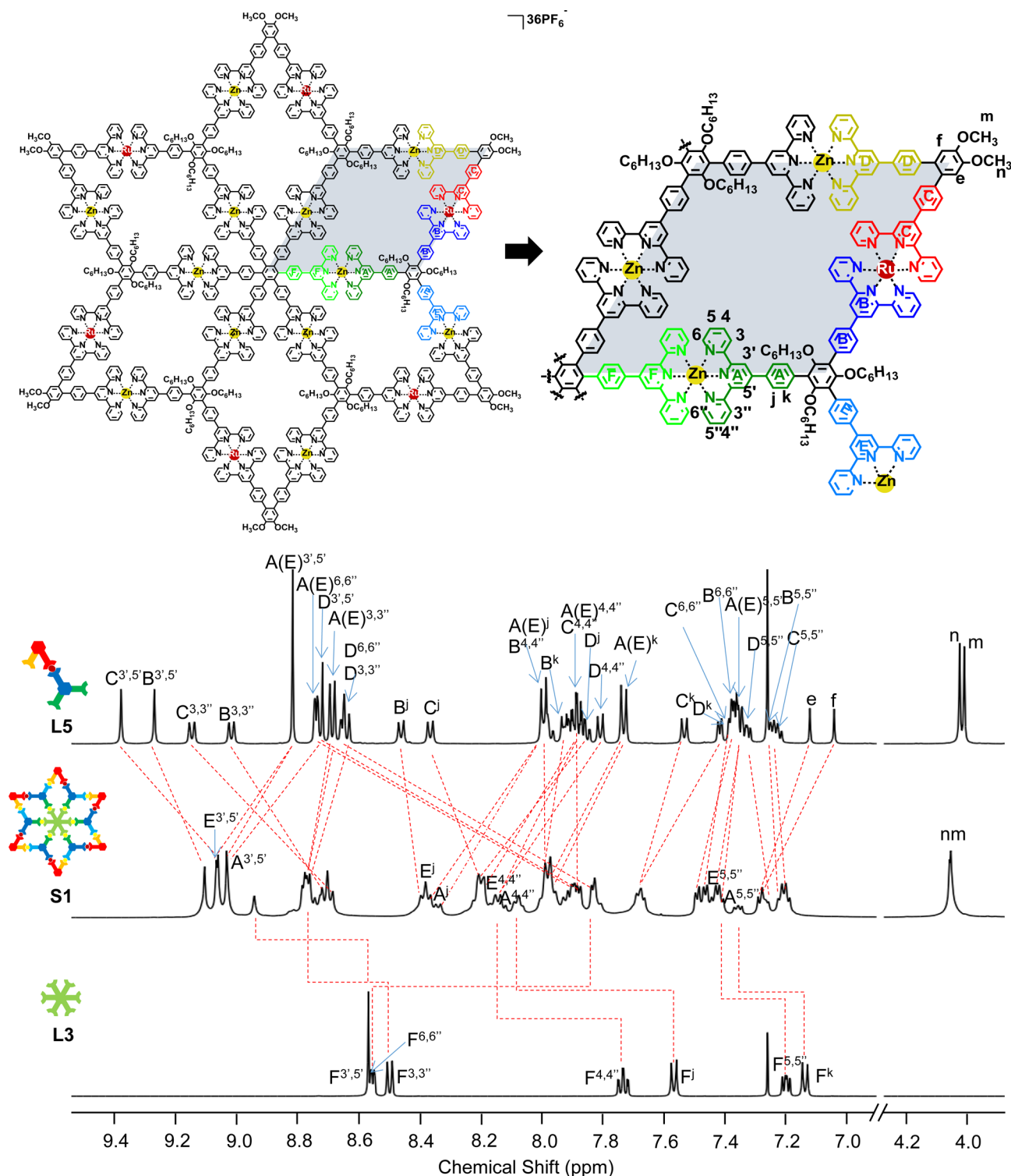


Figure 1. ^1H NMR spectra (500 MHz) of **L5** and **L3** in CDCl_3 and **S1** in CD_3CN .

the D-tpy group with $\text{Zn}(\text{II})$ to form a unique structure. In the alkyl region, the overlapped signals of the terminal methyl on the hexyloxy chain of **L5** were split into two sets of triplet peaks with an integration ratio of 2:1 (Figure S42, [Supporting Information](#)). This indicated the formation of the designed snowflake structure, in which two of the three long alkyl chains on each arm were settled inside the snowflake structure, while the third one stayed freely outside the structure. All the assignments shown in [Figure 1](#) are readily confirmed by the 2D

COSY (correlation spectroscopy), 2D NOESY (nuclear Overhauser effect spectroscopy), and 2D ROESY (rotating-frame Overhauser spectroscopy) spectra (Figures S45–S48, S56, and S57, [Supporting Information](#)).

Diffusion-ordered NMR spectroscopy (DOSY) provides dimensional information on **S1**, as shown in [Figure 2a](#) (full-ranged spectra are shown in Figure S62, [Supporting Information](#)). The spectrum shows a narrow band of signal with a diffusion coefficient (D) around $1.70 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$.

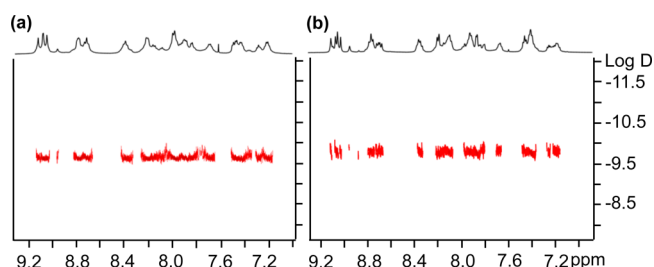


Figure 2. 2D DOSY (400 MHz, CD₃CN, 300 K) spectra of (a) S1 and (b) S2.

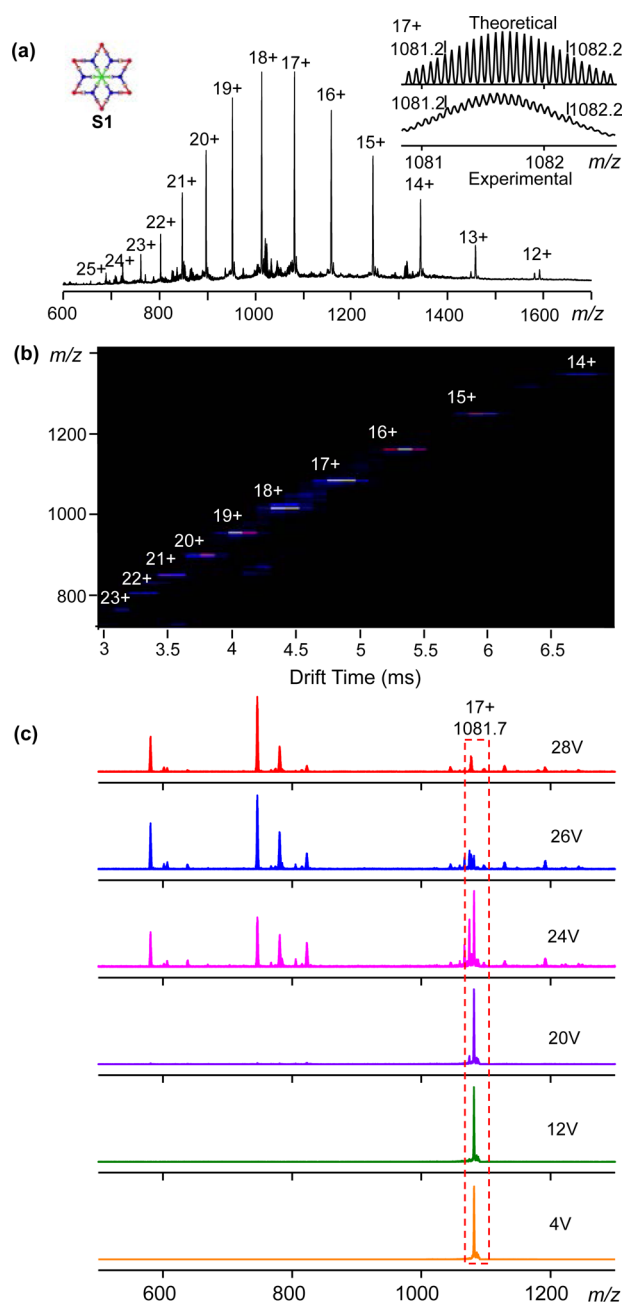


Figure 3. (a) ESI-MS and (b) TWIM-MS plots (m/z vs drift time) of S1. (c) gMS² of S1 at m/z 1081.7 with different collision energies.

Using the modified Stocks–Einstein equation based on the oblate spheroid model (the detailed calculation procedure is summarized in the [Supporting Information](#)),⁴¹ the experimental

radius, i.e., the semimajor axis radius, of S1 was found to be 4.90 nm, which is comparable to the outer radii measured from the molecular modeling of S1 (~4.3 nm, [Figure 4](#)) with a snowflake shape.

Multidimensional mass spectrometry, including ESI-MS, traveling wave ion mobility mass spectrometry (TWIM-MS),⁴² and gradient tandem mass spectrometry (gMS²),⁴³ provide further evidence for the formation of the accepted discrete assembly. One dominant set of peaks with continuous charge states from 12+ to 25+ was observed in ESI-MS ([Figure 3a](#)), due to successive loss of the counterions, PF₆[−]. After deconvolution of m/z , the average measured molecular mass of the assembly is 20830 Da, which exactly agrees with the molecular composition of S1 [(C₁₃₇H₁₁₇N₁₅O₅)₆(C₁₃₂H₈₄N₁₈)Zn₁₂Ru₆(PF₆)₃₆]. The experimental isotope patterns of each charge state ([Figure S2](#), [Supporting Information](#)) are consistent with the theoretical simulations based on the chemical composition of S1. We also conducted self-assembly of L5 and L3 with slightly different ratios of ligands and metals ([Figure S67](#), [Supporting Information](#)). ESI-MS showed that S1 could even be assembled not in the perfect stoichiometry. TWIM-MS results ([Figure 3b](#)) show a single set of signals with a narrow distribution at each charge state (m/z), which unambiguously addresses that no other isomers or conformers of S1 existed. It is suggested that the snowflake is more rigid and shape-persistent compared to the previous dendron-functionalized macrocycles.^{36,37}

TWIM-MS exhibits more information, i.e., collision cross sections (CCSs), on the shape and size of the assemblies. CCS data of S1 on each charge state are summarized in [Table 1](#), with an average value of 2959.1 Å². For comparison, the theoretical CCSs of S1 were calculated on the basis of the 70 candidate structures obtained from each annealing cycle of the energy-minimized molecular model of S1. The averaged result of the calculated CCSs is 3054.6 Å² (by the trajectory method),⁴⁴ which is in good agreement with the experimental one, further confirming the snowflake shape of S1. gMS² was also employed to measure the stability of the supramolecular complex. [Figure 3c](#) displays the gMS² results of the 17+ ions (m/z 1081.7) of S1 by applying a graduated increase of the collision energy from 4 V. No significant fragment peaks were observed under 20 V. All the ions were completely dissociated at 28 V, which gives a center-of-mass collision energy of 0.06 eV and is used for the comparison with snowflake S2.

Transmission electron microscopy (TEM) was employed to further confirm the shape and size of snowflake S1 ([Figure 4c](#)). High-resolution TEM imaging clearly displayed star-shaped hollow structures for S1 with size comparable to that of the modeling structures. Very interestingly, these supramolecules may further self-assemble into a 2D array with a uniform pattern on the lacey carbon-coated Cu grid. The measured diameter of these patterns from TEM (8.6 nm) was comparable to the molecular modeling diameter (8.5 nm) of S1. Furthermore, energy-dispersive X-ray spectroscopy (EDXS) on the TEM patterns ([Figure S73](#), [Supporting Information](#)) showed N, O, Ru, and Zn signals, confirming that the star-shaped pattern observed in TEM was derived from metallo-supramolecules. TEM imaging on a regular carbon-coated Cu grid only showed individual dots without a hollow pattern, perhaps due to the background signals from the thick layer of carbon ([Figure S72](#), [Supporting Information](#)). We also performed TEM for other negative controls with nonideal

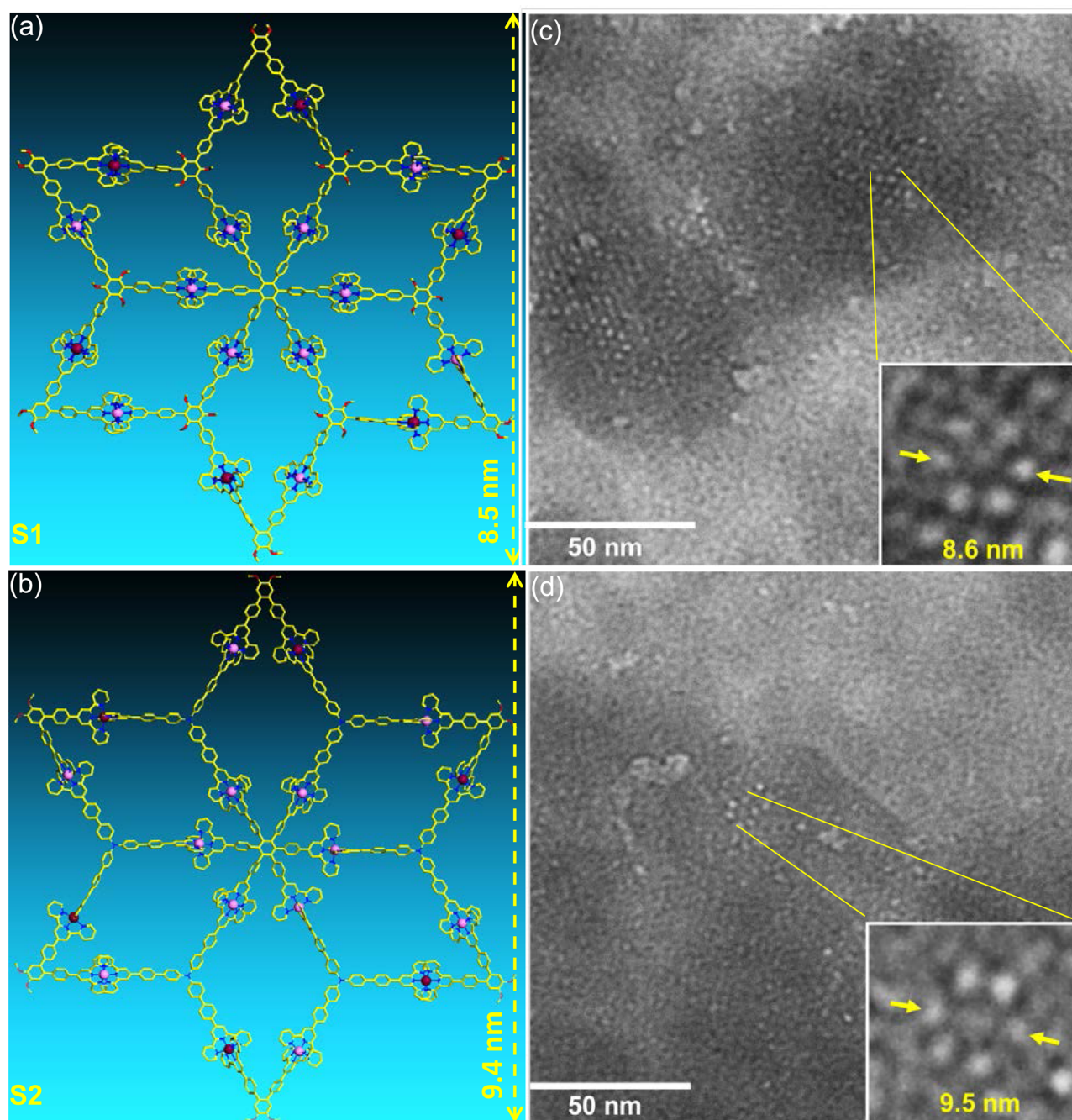


Figure 4. (a, b) Representative energy-minimized structures from molecular modeling of S1 and S2. (c, d) TEM images of S1 and S2. Alkyl chains were omitted for clarity.

ratios of ligands and metals (Figure S69, [Supporting Information](#)), but did not observe any snowflake-like pattern.

We also performed atomic force microscopy (AFM) and scanning tunneling microscopy (STM) imaging for S1. In the AFM measurement, we observed individual dots with a single molecular height, which is consistent with our molecular modeling (Figure S75a,b, [Supporting Information](#)). Note that the measured width of the molecule was much larger than the height measured by AFM and molecular modeling because of an inevitable tip broadening effect.^{28f} In the STM measurement (Figure S76a,b, [Supporting Information](#)), the samples were applied to highly oriented pyrolytic graphite (HOPG) by drop-casting. Instead of observing individual supramolecules, STM showed ribbon-like nanostructures with pores. We speculated

that formation of these nanoribbons was driven by the interaction between metallo-supramolecules and the HOPG surface along with intermolecular interactions. This is consistent with our recent study of hierarchical self-assembly of supramolecular metal–organic nanoribbons by supramolecular concentric hexagons.^{28g}

Synthesis, Self-Assembly, and Characterization of Supersnowflake 2 (S2). Encouraged by the success of S1, Ru(II)–organic ligand L6 with a similar geometry but slightly different length was designed for the self-assembly of snowflake S2 with L3 ([Scheme 1](#)). The major difference is the phenyl group in L5 was replaced by a triphenylamine moiety in L6. With such a design, we might be able to assemble a larger version of supersnowflake S2 with a diameter around 9.4 nm

Table 1. Experimental and Theoretical Collision Cross Sections (CCSs)

	drift time (ms)	CCS ($\text{\AA}^2$)	CCS av ($\text{\AA}^2$)	CCS (calcd av) ($\text{\AA}^2$)
S1	6.50 (14+)	2929.5	2959.1 $\pm$ 48.9	3054.6 $\pm$ 64.2
	5.73 (15+)	2917.3		
	5.18 (16+)	2933.9		
	4.63 (17+)	2918.5		
	4.19 (18+)	2912.9		
	3.86 (19+)	2928.3		
	3.64 (20+)	2976.5		
	3.42 (21+)	3010.5		
	3.20 (22+)	3029.7		
	2.98 (23+)	3033.6		
	6.17 (15+)	3047.1	3059.0 $\pm$ 18.2	3176.5 $\pm$ 57.1
S2	5.51 (16+)	3043.4		
	4.96 (17+)	3041.8		
	4.52 (18+)	3047.7		
	4.19 (19+)	3077.3		
	3.86 (20+)	3085.2		
	3.53 (21+)	3070.7		

according to molecular modeling (Figure 4b), in which no significant structural distortion was observed. We expected that this robust snowflake system would readily accommodate the variation of the ligand in the outer rim but without losing the entire structural integrity benefiting from the energy-favorable features of the structure.

A synthetic strategy similar to that of **L5** was carried out to prepare Ru(II)–organic ligand **L6** by using tris(4-bromophenyl)amine as the starting material (Scheme S4, Supporting Information), and the subsequent assembly of **S2** was performed by mixing **L6**, **L3**, and $\text{Zn}(\text{NO}_3)_2$ in the stoichiometric ratio (6:1:12). The formation of the desired structure was first characterized by ^1H NMR (Figure 5), and the corresponding 2D NMR results are summarized in the Supporting Information. The aromatic region in the spectrum of **S2** is more complicated than that in the spectrum of **S1**, because of the new induction of three sets of phenyl protons. Nevertheless, relatively sharp peaks and characteristic shifts of $\text{tpyH}^{3',5'}$ (downfield), $\text{tpyH}^{6,6''}$ (upfield), and methoxyl protons (merged into one peak) after the assembly together suggested the formation of a unique structure via the expected tpy –Zn interaction.

The ESI-MS result of **S2** (Figure 6a) displays a series of peaks with continuous charge states from 10+ to 20+, and each charge state displays well-resolved isotope patterns, which are consistent with the theoretical ones (Figure S3, Supporting Information). The averaged measured mass of **S2** was 20030 Da, supporting its composition $[(\text{C}_{131}\text{H}_{90}\text{N}_{16}\text{O}_2)_6(\text{C}_{132}\text{H}_{84}\text{N}_{18})\text{Zn}_{12}\text{Ru}_6(\text{PF}_6)_{36}]$. Narrowly distributed signals of each charge state are also exhibited in TWIM-MS, indicating its exceptional rigidity and shape persistence. However, **S2** was not observed in ESI-MS for self-assembly of **L6** and **L3** with slightly different ratios of ligands and metals (Figure S68, Supporting Information).

Experimental CCSs obtained from the TWIM-MS results of **S2** were summarized in Table 1. The averaged CCS is 3059 $\text{\AA}^2$, consistent with the averaged theoretical result (3176 $\text{\AA}^2$). The increment of the CCS value of **S2** from **S1** comes from the elongated outer rims of the snowflake as we anticipated in the design. Additionally, 2D DOSY of **S2** shows a lower value of the diffusion efficiency (*D*) (Figure 2b; full-ranged spectra are

shown in Figure S63, Supporting Information) with a higher conducted semimajor axis radius (5.9 nm), and TEM imaging of **S2** also displays clear images of snowflake-shaped patterns with a narrow dispersity of the diameter around 9.5 nm (Figure 4d). TEM was also conducted for negative controls with nonideal ratios of ligands and metals (Figure S70, Supporting Information), but did not display any snowflake-like pattern. As for **S1**, similar AFM and STM results were obtained for **S2** (Figure S75c-d, S76c-d, Supporting Information).

Further characterization of **S2** via its stability was performed by gMS^2 , shown in Figure 6c. The 17+ ions ($m/z = 1034.7$) completely disappeared under 24 V, which was 4 V lower than the same situation for **S1**. This suggested a lower stability of **S2** than that of **S1**. The lower stability of **S2** was determined by its larger size and lower density of coordination sites (DOCS),^{28c,45} accordingly. DOCS provides the quantitative relationship between the stability and coordination site number/size (theoretical CCSs) of the assemblies. With the same coordination site number, a larger size brings about a lower DOCS and, thus, a lower stability of the assembly structure. The DOCS value of **S1** is 0.0118 site/ $\text{\AA}^2$, which is 5% larger than that of **S2** (0.0113 site/ $\text{\AA}^2$).

Intramolecular Dynamic Exchange within **S1** and **S2**.

In both snowflakes **S1** and **S2**, there are two sets of tpy, i.e., A and E, from ligands **L5** and **L6**. In the scaffolds of the snowflakes, a slow intramolecular dynamic exchange between A- and E-tpy may occur in **S1** and **S2** alone. Therefore, we performed a detailed ROESY NMR study with a variety of mixing times to address the possible intramolecular exchange. As shown in Figures S60 and S61 (Supporting Information), the cross-exchange signals of A- and E-tpy can be observed at mixing times over 200 ms, which indicated a very slow exchange process at 300 K compared to those of other reported compounds with mixing times of less than 100 ms.⁴⁶

Variable-temperature ^1H NMR was also carried out to investigate the dynamic exchange. As shown in Figure S65 (Supporting Information), the signals of the 3',5'-protons of the A- and E-tpy groups of **S1** were well-split at 300 K with a 0.034 ppm distance, which was gradually decreased to 0.027 ppm by increasing the operating temperature to 340 K (65 $^\circ\text{C}$). This indicates that the dynamic exchange of A- and E-tpy can be enhanced by increasing the temperature.⁴⁷ Similar emerging signals of A3',5' and E3',5' of **S2** were also recorded with variable-temperature ^1H NMR spectroscopy (Figure S66, Supporting Information).

Intermolecular Dynamic Exchange between **S1** and **S2**.

In spite of the difference of ligands **L5** and **L6** in the outer rim, both **S1** and **S2** could be assembled with similar structures. If preassembled **S1** and **S2** were mixed in solution, the following three circumstances could occur: coexistence of two discrete supersnowflakes without communication, collapse and formation of unidentified complexes, or generation of hybrid snowflakes through ligand exchange. To answer the above question, preassembled **S1** and **S2** (CH_3CN , 1 mg/mL) solutions were mixed in a 1:1 ratio at room temperature for a dynamic behavior study. The entire process was monitored by ESI-MS and TWIM-MS (Figure 7).

At 0 min after mixing, ESI-MS displayed two series of peaks corresponding to **S1** and **S2**, respectively. Remarkably, 2D separation by TWIM-MS clearly showed two sets of signals without any superimposition (Figures 7a,b). The slow dynamic ligand exchange started around 5 h after the mixing, indicating the great stability of each snowflake. In addition to the original

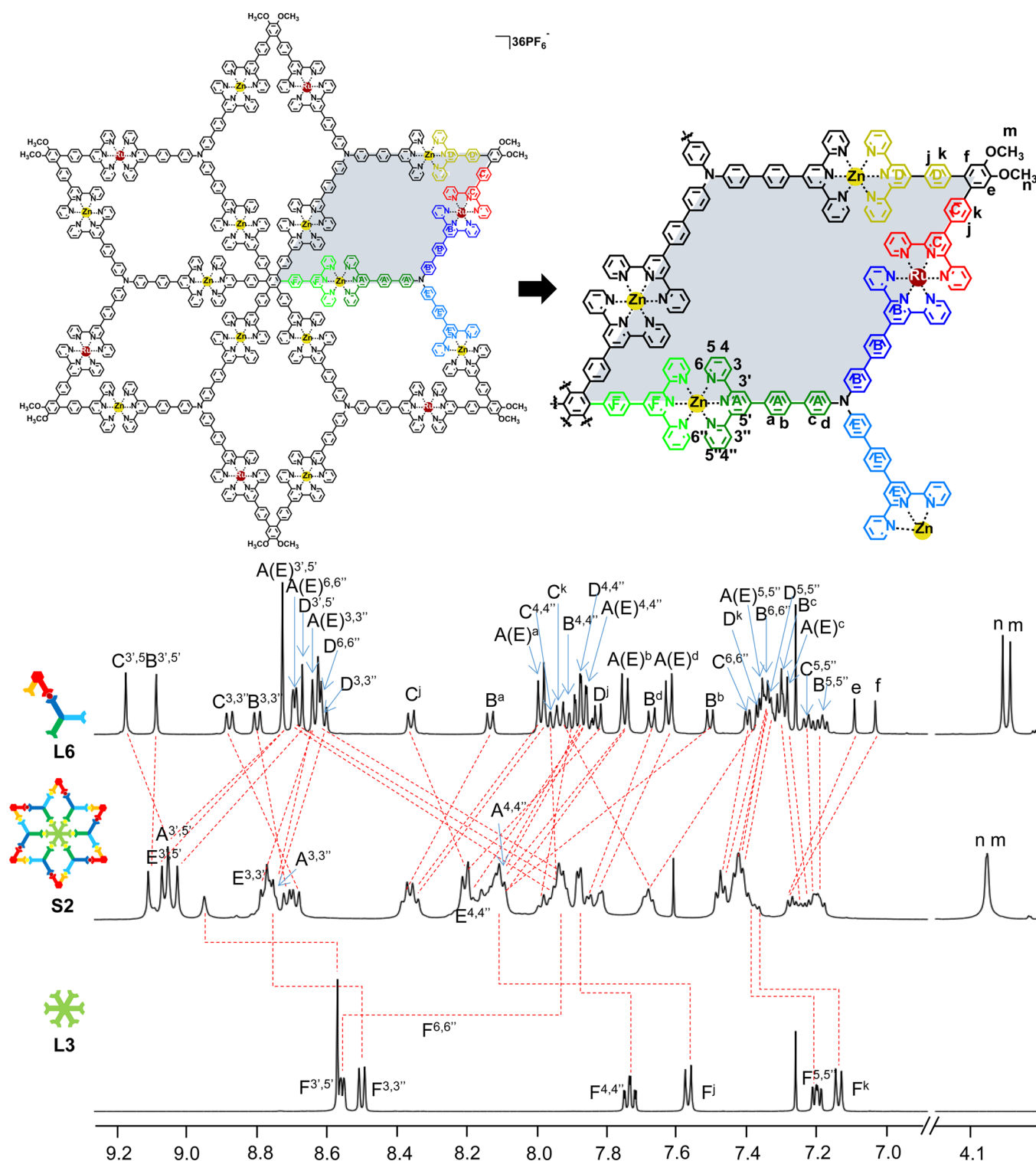


Figure 5. ^1H NMR spectra (500 MHz) of **L6** and **L3** in CDCl_3 and **S2** in CD_3CN .

peaks of **S1** and **S2**, for instance, **S1** $^{16+}$ and **S2** $^{16+}$, five new peaks appeared, corresponding to all the statistical hybrid snowflakes. After 5 d, no significant change was detected by mass spectrometry. In the 2D spectrum of TWIM-MS (Figure 7d), all the hybrid signals drifted out with very similar times, suggesting they should possess snowflake architectures similar to those of **S1** and **S2**. We speculated that the dynamic exchange reached equilibrium around 5 d with partial **S1** and **S2** left. To validate this assumption, a control study was

performed by mixing equal molar amounts of **L5** and **L6** with **L3** and Zn(II) in a 3:3:1:12 ratio. After heating at 50°C for 15 h, the mass spectrometry exhibited a result almost identical to that of the 5 d dynamic exchange (Figure S7, Supporting Information). Finally, we performed DOSY-NMR for the mixtures of snowflakes with a high number of possible structures and isomers after complete intermolecular dynamic exchange. The measured D of the mixture was between those of **S1** and **S2** (Figure S64, Supporting Information), indicating

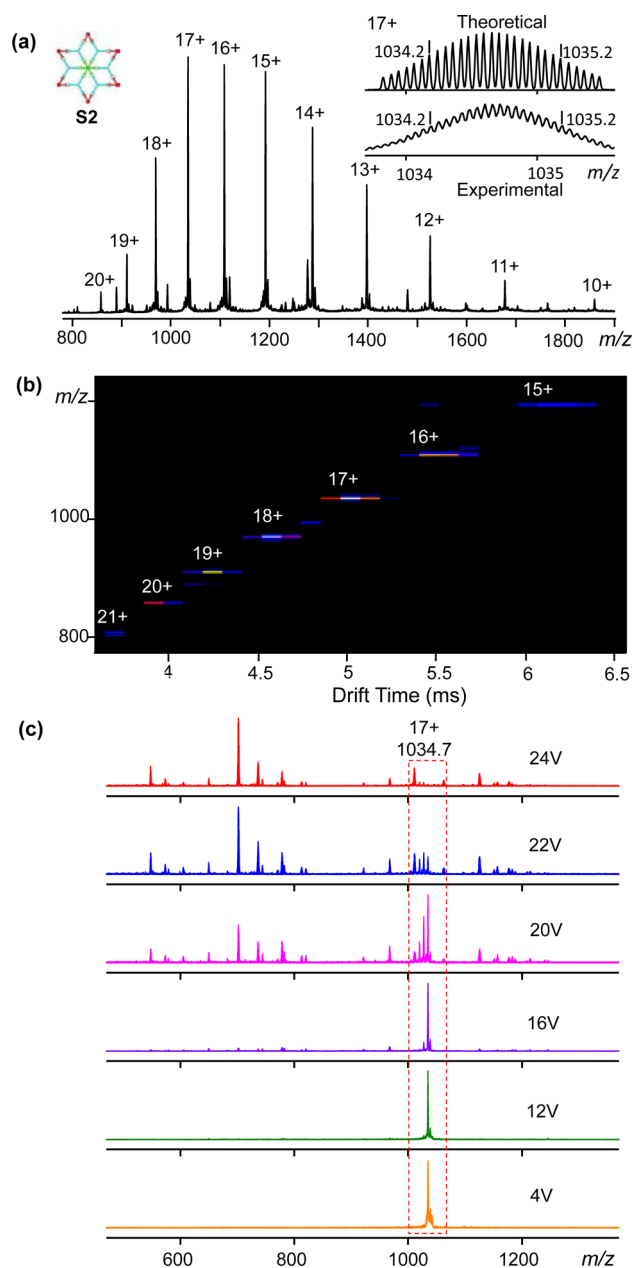


Figure 6. (a) ESI-MS and (b) TWIM-MS plots of **S2**. (c) gMS² of 17+ ions of **S2** at m/z 1034.7 with different collision energies.

that those hybrid supramolecular snowflakes have an intermediate size between those of **S1** and **S2**.

Physical Properties. Considering the tpy and triphenylamine have been widely used in optoelectronics in both photoreceptor devices and organic light-emitting diodes,⁴⁸ we performed the following photo- and electrochemical properties study: In the absorption spectra (Figure 8a), both **S1** and **S2** show characteristic absorption bands peaking at 495 nm, corresponding to allowed metal-to-ligand charge transfer (MLCT) transitions.⁴⁹ **S2** also exhibits a unique broad absorption band located around 400 nm, which comes from intramolecular charge transfer (ICT) transition from the triphenylamine moiety to the tpy–M complex moieties.⁴⁹ The emissions of **S1** and **S2** were detected in CH₃CN solution under 73 K (Figure 8b), showing peaks at 640 nm (**S1**) and 650 nm (**S2**).

The electrochemical properties of these two complexes in DMF were studied, and the results are shown in Figure 9. Both **S1** and **S2** exhibit three tpy-ligand-centered redox couples between –2.0 and –0.5 V. Between 1.0 and 1.5 V, **S2** showed two irreversible oxidation peaks, corresponding to the oxidation of the triphenylamine moiety and Ru(II) atoms, while **S1** displayed only one oxidation peak of Ru(II). In addition, under the effect of the triphenylamine group, although the peak-to-peak potential difference for each redox couple was still very similar between the two complexes, the potentials of each single peak of **S2** were moved to more positive values than those of **S1**.

CONCLUSIONS

By utilizing the diverse coordination ability of tpy with transition metals, we have designed and assembled two snowflake supramolecules using the combination of Ru(II) and Zn(II). Instead of using direct self-assembly, such giant metallo-supramolecules were approached through a stepwise synthesis strategy, which successfully prevented the formation of undesired assemblies by the self-sorting of individual ligands. For the first time, in the tpy-based supramolecular field, we were able to construct complex supramolecular architectures with three types of organic ligands. Additionally, dynamic ligand exchange of two preassembled architectures indicated that this versatile system was capable of accommodating the variation of the outer rims to form a series of hybrid snowflakes but without the entire structural integrity. We anticipate that these discrete and shape-persistent supramolecular architectures may serve as a model system for further study of the self-assembly behavior and physical properties of 2D materials.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/jacs.7b01326.

Synthetic details, molecular modeling, and ligand and complex characterization, including ¹H NMR, ¹³C NMR, 2D COSY, 2D NOESY, 2D ROESY, ESI-MS, TWIM-MS, and MALDI-TOF (PDF)

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Notes

The authors declare no competing financial interest.

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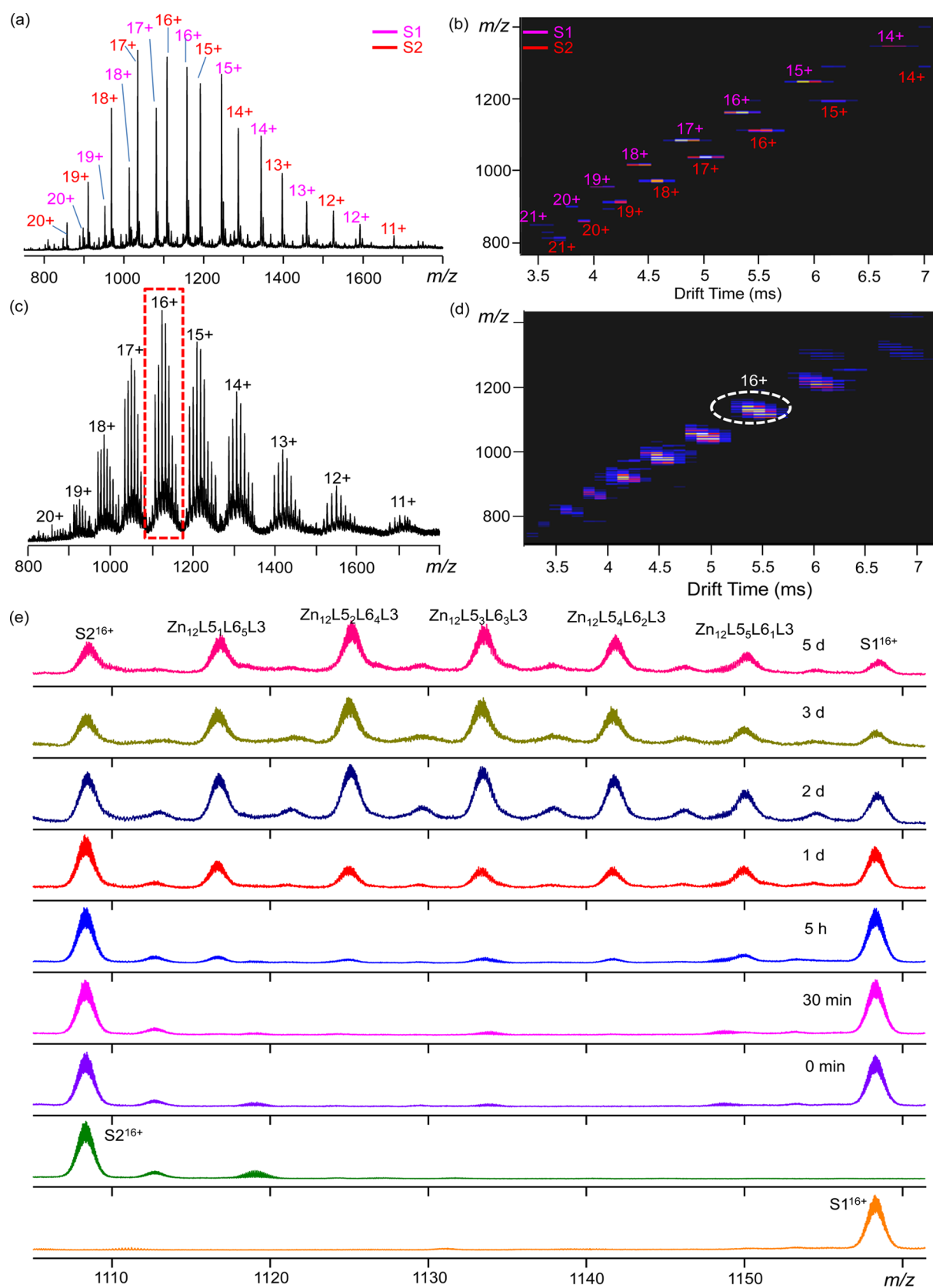


Figure 7. Dynamic exchange of preassembled S1 and S2. (a) ESI-MS and (b) TWIM-MS plots of preassembled S1 and S2 at 0 min. (c) ESI-MS and (d) TWIM-MS plots of preassembled S1 and S2 at 5 d. (e) Time-dependent spectra of dynamic exchange of S1 and S2 at 0 min, 30 min, 5 h, 1 d, 2 d, 3 d, and 5 d for an expanded region of 16+ signals.

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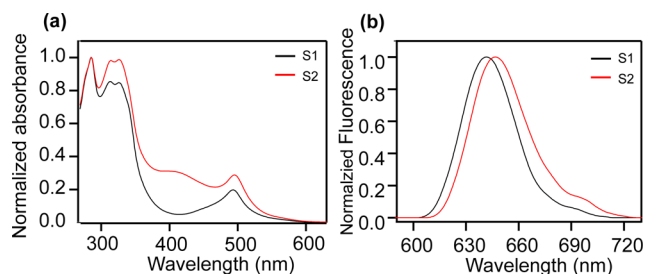


Figure 8. UV-vis (a) (10^{-6} M in CH_3CN , room temperature) and emission (b) (10^{-6} M in CH_3CN , 73 K) of complexes of S1 and S2.

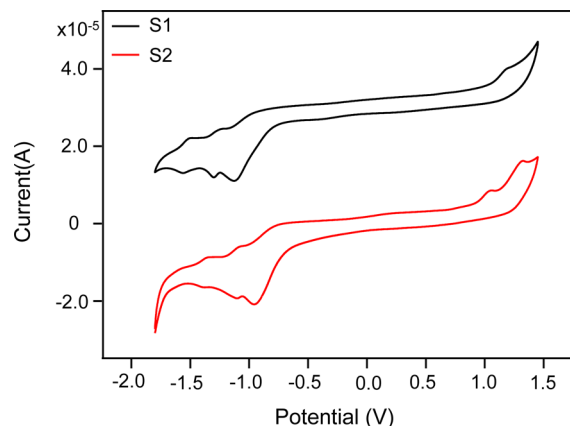


Figure 9. Cyclic voltammograms of solutions of S1 and S2 (performed in 0.1 M $n\text{-Bu}_4\text{NBF}_4$ in DMF at 298 K).

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REFERENCES

- (1) (a) Cram, D. J. *Angew. Chem., Int. Ed. Engl.* **1988**, *27*, 1009. (b) Lehn, J.-M. *Angew. Chem., Int. Ed. Engl.* **1988**, *27*, 89. (c) Lehn, J.-M. *Science* **2002**, *295*, 2400. (d) Pedersen, C. J. *Angew. Chem., Int. Ed. Engl.* **1988**, *27*, 1021.
- (2) (a) Sauvage, J.-P.; Collin, J.-P.; Chambron, J.-C.; Guillerez, S.; Coudret, C.; Balzani, V.; Barigelli, F.; De Cola, L.; Flamigni, L. *Chem. Rev.* **1994**, *94*, 993. (b) Holliday, B. J.; Mirkin, C. A. *Angew. Chem., Int. Ed.* **2001**, *40*, 2022. (c) Davis, A. V.; Yeh, R. M.; Raymond, K. N. *Proc. Natl. Acad. Sci. U. S. A.* **2002**, *99*, 4793. (d) Dalgarno, S. J.; Power, N. P.; Atwood, J. L. *Coord. Chem. Rev.* **2008**, *252*, 825. (e) Smulders, M. M.; Riddell, I. A.; Browne, C.; Nitschke, J. R. *Chem. Soc. Rev.* **2013**, *42*, 1728. (f) Ward, M. D.; Raithby, P. R. *Chem. Soc. Rev.* **2013**, *42*, 1619. (g) Saha, M. L.; De, S.; Pramanik, S.; Schmittle, M. *Chem. Soc. Rev.* **2013**, *42*, 6860. (h) Han, M.; Engelhard, D. M.; Clever, G. H. *Chem. Soc. Rev.* **2014**, *43*, 1848.
- (3) (a) Chakrabarty, R.; Mukherjee, P. S.; Stang, P. J. *Chem. Rev.* **2011**, *111*, 6810. (b) Cook, T. R.; Stang, P. J. *Chem. Rev.* **2015**, *115*, 7001. (c) Leininger, S.; Olenyuk, B.; Stang, P. J. *Chem. Rev.* **2000**, *100*, 853.
- (4) (a) Yoshizawa, M.; Klosterman, J. K.; Fujita, M. *Angew. Chem., Int. Ed.* **2009**, *48*, 3418. (b) Harris, K.; Fujita, D.; Fujita, M. *Chem. Commun.* **2013**, *49*, 6703.
- (5) (a) Hasenknopf, B.; Lehn, J.-M.; Boumediene, N.; Leize, E.; Van Dorsselaer, A. *Angew. Chem., Int. Ed.* **1998**, *37*, 3265. (b) Hasenknopf, B.; Lehn, J.-M.; Kneisel, B. O.; Baum, G.; Fenske, D. *Angew. Chem., Int. Ed. Engl.* **1996**, *35*, 1838. (c) Hasenknopf, B.; Lehn, J.-M.; Boumediene, N.; Dupont-Gervais, A.; Van Dorsselaer, A.; Kneisel, B.; Fenske, D. *J. Am. Chem. Soc.* **1997**, *119*, 10956. (d) Ayme, J.-F.; Beves, J. E.; Campbell, C. J.; Leigh, D. A. *Angew. Chem., Int. Ed.* **2014**, *53*, 7823. (e) Sorensen, A.; Castilla, A. M.; Ronson, T. K.; Pittelkow, M.; Nitschke, J. R. *Angew. Chem., Int. Ed.* **2013**, *52*, 11273.
- (6) (a) Bassani, D. M.; Lehn, J.-M.; Fromm, K.; Fenske, D. *Angew. Chem., Int. Ed.* **1998**, *37*, 2364. (b) Barboiu, M.; Vaughan, G.; Graff, R.; Lehn, J.-M. *J. Am. Chem. Soc.* **2003**, *125*, 10257. (c) Toyota, S.; Woods, C. R.; Benaglia, M.; Haldimann, R.; Wärnmark, K.; Hardcastle, K.; Siegel, J. S. *Angew. Chem., Int. Ed.* **2001**, *40*, 751.
- (7) (a) Fujita, M.; Yazaki, J.; Ogura, K. *J. Am. Chem. Soc.* **1990**, *112*, 5645. (b) Fujita, M.; Nagao, S.; Iida, M.; Ogata, K.; Ogura, K. *J. Am. Chem. Soc.* **1993**, *115*, 1574.
- (8) (a) Stang, P. J.; Cao, D. H. *J. Am. Chem. Soc.* **1994**, *116*, 4981. (b) Stang, P. J.; Cao, D. H.; Saito, S.; Arif, A. M. *J. Am. Chem. Soc.* **1995**, *117*, 6273. (c) Stang, P. J.; Chen, K. *J. Am. Chem. Soc.* **1995**, *117*, 1667. (d) Manna, J.; Whiteford, J. A.; Stang, P. J.; Muddiman, D. C.; Smith, R. D. *J. Am. Chem. Soc.* **1996**, *118*, 8731. (e) Stang, P. J.; Persky, N. E.; Manna, J. *J. Am. Chem. Soc.* **1997**, *119*, 4777. (f) Ghosh, K.; Yang, H.-B.; Northrop, B. H.; Lyndon, M. M.; Zheng, Y.-R.; Muddiman, D. C.; Stang, P. J. *J. Am. Chem. Soc.* **2008**, *130*, 5320. (g) Chakrabarty, R.; Stang, P. J. *J. Am. Chem. Soc.* **2012**, *134*, 14738.
- (9) (a) Chan, Y.-T.; Li, X.; Yu, J.; Carri, G. A.; Moorefield, C. N.; Newkome, G. R.; Wesdemiotis, C. *J. Am. Chem. Soc.* **2011**, *133*, 11967. (b) Sautter, A.; Schmid, D. G.; Jung, G.; Würthner, F. *J. Am. Chem. Soc.* **2001**, *123*, 5424. (c) Sautter, A.; Kaletas, B. K.; Schmid, D. G.; Dobrawa, R.; Zimine, M.; Jung, G.; van Stokkum, I. H.; De Cola, L.; Williams, R. M.; Würthner, F. *J. Am. Chem. Soc.* **2005**, *127*, 6719. (d) You, C.-C.; Würthner, F. *J. Am. Chem. Soc.* **2003**, *125*, 9716. (e) Jeong, K. S.; Kim, S. Y.; Shin, U.-S.; Kogej, M.; Hai, N. T.; Broekmann, P.; Jeong, N.; Kirchner, B.; Reiher, M.; Schalley, C. A. *J. Am. Chem. Soc.* **2005**, *127*, 17672. (f) Safarowsky, C.; Merz, L.; Rang, A.; Broekmann, P.; Hermann, B.; Schalley, C. A. *Angew. Chem., Int. Ed.* **2004**, *43*, 1291. (g) Campos-Fernández, C. S.; Clérac, R.; Koomen, J. M.; Russell, D. H.; Dunbar, K. R. *J. Am. Chem. Soc.* **2001**, *123*, 773. (h) Campos-Fernández, C. S.; Schottel, B. L.; Chifotides, H. T.; Bera, J. K.; Bacsá, J.; Koomen, J. M.; Russell, D. H.; Dunbar, K. R. *J. Am. Chem. Soc.* **2005**, *127*, 12909. (i) Chifotides, H. T.; Giles, I. D.; Dunbar, K. R. *J. Am. Chem. Soc.* **2013**, *135*, 3039.
- (10) (a) Sun, Q.-F.; Iwasa, J.; Ogawa, D.; Ishido, Y.; Sato, S.; Ozeki, T.; Sei, Y.; Yamaguchi, K.; Fujita, M. *Science* **2010**, *328*, 1144. (b) Fujita, D.; Ueda, Y.; Sato, S.; Yokoyama, H.; Mizuno, N.; Kumasaka, T.; Fujita, M. *Chem.* **2016**, *1*, 91. (c) Tominaga, M.; Suzuki, K.; Kawano, M.; Kusukawa, T.; Ozeki, T.; Sakamoto, S.; Yamaguchi, K.; Fujita, M. *Angew. Chem., Int. Ed.* **2004**, *43*, 5621.
- (11) (a) Yan, X.; Cook, T. R.; Wang, P.; Huang, F.; Stang, P. J. *Nat. Chem.* **2015**, *7*, 342. (b) Olenyuk, B.; Whiteford, J. A.; Fechtenkotter, A.; Stang, P. J. *Nature* **1999**, *398*, 796. (c) Kuehl, C. J.; Kryshenko, Y. K.; Radhakrishnan, U.; Seidel, S. R.; Huang, S. D.; Stang, P. J. *Proc. Natl. Acad. Sci. U. S. A.* **2002**, *99*, 4932. (d) Leininger, S.; Fan, J.; Schmitz, M.; Stang, P. J. *Proc. Natl. Acad. Sci. U. S. A.* **2000**, *97*, 1380. (e) Olenyuk, B.; Levin, M. D.; Whiteford, J. A.; Shield, J. E.; Stang, P. J. *J. Am. Chem. Soc.* **1999**, *121*, 10434.
- (12) (a) Wang, M.; Wang, C.; Hao, X.-Q.; Li, X.; Vaughn, T. J.; Zhang, Y.-Y.; Yu, Y.; Li, Z.-Y.; Song, M.-P.; Yang, H.-B.; Li, X. *J. Am. Chem. Soc.* **2014**, *136*, 10499. (b) Xie, T.-Z.; Liao, S.-Y.; Guo, K.; Lu, X.; Dong, X.; Huang, M.; Moorefield, C. N.; Cheng, S. Z.; Liu, X.; Wesdemiotis, C.; Newkome, G. R. *J. Am. Chem. Soc.* **2014**, *136*, 8165. (c) Xie, T. Z.; Guo, K.; Guo, Z.; Gao, W. Y.; Wojtas, L.; Ning, G. H.; Huang, M.; Lu, X.; Li, J. Y.; Liao, S. Y.; Chen, Y.-S.; Moorefield, C. N.; Saunders, M. J.; Cheng, S. Z. D.; Wesdemiotis, C.; Newkome, G. R. *Angew. Chem.* **2015**, *127*, 9356. (d) Hiraoka, S.; Harano, K.; Shiro, M.; Ozawa, Y.; Yasuda, N.; Toriumi, K.; Shionoya, M. *Angew. Chem.* **2006**, *118*, 6638. (e) Clever, G. H.; Tashiro, S.; Shionoya, M. *J. Am. Chem. Soc.* **2010**, *132*, 9973. (f) Bar, A. K.; Chakrabarty, R.; Mostafa, G.; Mukherjee, P. S. *Angew. Chem., Int. Ed.* **2008**, *47*, 8455. (g) Han, M.; Luo, Y.; Damaschke, B.; Gómez, L.; Ribas, X.; Jose, A.; Peretzki, P.; Seibt, M.; Clever, G. H. *Angew. Chem., Int. Ed.* **2016**, *55*, 445. (h) Riddell, I. A.; Smulders, M. M.; Clegg, J. K.; Hristova, Y. R.; Breiner, B.; Thoburn, J. D.; Nitschke, J. R. *Nat. Chem.* **2012**, *4*, 751. (i) Zhang, H.; Lee, J.; Lammer, A. D.; Chi, X.; Brewster, J. T.; Lynch, V. M.; Li, H.; Zhang, Z.; Sessler, L. J. *J. Am. Chem. Soc.* **2016**, *138*, 4573.

- (13) (a) Kraus, T.; Buděšínský, M.; Cvačka, J.; Sauvage, J. P. *Angew. Chem., Int. Ed.* **2005**, *45*, 258. (b) Beyler, M.; Heitz, V.; Sauvage, J.-P. *J. Am. Chem. Soc.* **2010**, *132*, 4409. (c) Joosten, A.; Trolez, Y.; Collin, J.-P.; Heitz, V. r.; Sauvage, J.-P. *J. Am. Chem. Soc.* **2012**, *134*, 1802. (d) Yang, Y.-D.; Fan, C.-C.; Rambo, B. M.; Gong, H.-Y.; Xu, L.-J.; Xiang, J.-F.; Sessler, J. L. *J. Am. Chem. Soc.* **2015**, *137*, 12966.
- (14) (a) Dietrich-Buchecker, C. O.; Sauvage, J. P.; Kern, J. M. *J. Am. Chem. Soc.* **1984**, *106*, 3043. (b) Dietrich-Buchecker, C. O.; Sauvage, J.; Kintzinger, J. *Tetrahedron Lett.* **1983**, *24*, 5095.
- (15) Fujita, M.; Ibukuro, F.; Hagihara, H.; Ogura, K. *Nature* **1994**, *367*, 720.
- (16) (a) Schulze, M.; Kunz, V.; Frischmann, P. D.; Würthner, F. *Nat. Chem.* **2016**, *8*, 576. (b) Zhu, R.; Lübben, J.; Dittrich, B.; Clever, G. H. *Angew. Chem., Int. Ed.* **2015**, *54*, 2796. (c) Freye, S.; Michel, R.; Stalke, D.; Pawliczek, M.; Frauendorf, H.; Clever, G. H. *J. Am. Chem. Soc.* **2013**, *135*, 8476. (d) Frank, M.; Hey, J.; Balcioglu, I.; Chen, Y. S.; Stalke, D.; Suenobu, T.; Fukuzumi, S.; Frauendorf, H.; Clever, G. H. *Angew. Chem., Int. Ed.* **2013**, *52*, 10102. (e) Freye, S.; Hey, J.; Torras-Galán, A.; Stalke, D.; Herbst-Irmer, R.; John, M.; Clever, G. H. *Angew. Chem., Int. Ed.* **2012**, *51*, 2191.
- (17) (a) Padilla-Tosta, M. E.; Fox, O. D.; Drew, M. G.; Beer, P. D. *Angew. Chem., Int. Ed.* **2001**, *40*, 4235. (b) Samanta, D.; Mukherjee, P. S. *J. Am. Chem. Soc.* **2014**, *136*, 17006. (c) Wood, C. S.; Ronson, T. K.; Belenguier, A. M.; Holstein, J. J.; Nitschke, J. R. *Nat. Chem.* **2015**, *7*, 354.
- (18) (a) Ayme, J.-F.; Beves, J. E.; Leigh, D. A.; McBurney, R. T.; Rissanen, K.; Schultz, D. J. *Am. Chem. Soc.* **2012**, *134*, 9488. (b) Ayme, J.-F.; Beves, J. E.; Leigh, D. A.; McBurney, R. T.; Rissanen, K.; Schultz, D. *Nat. Chem.* **2012**, *4*, 15. (c) Leigh, D. A.; Pritchard, R. G.; Stephens, A. J. *Nat. Chem.* **2014**, *6*, 978. (d) Guo, J.; Mayers, P. C.; Breault, G. A.; Hunter, C. A. *Nat. Chem.* **2010**, *2*, 218. (e) Barran, P. E.; Cole, H. L.; Goldup, S. M.; Leigh, D. A.; McGonigal, P. R.; Symes, M. D.; Wu, J.; Zengerle, M. *Angew. Chem., Int. Ed.* **2011**, *50*, 12280.
- (19) (a) Beves, J. E.; Campbell, C. J.; Leigh, D. A.; Pritchard, R. G. *Angew. Chem., Int. Ed.* **2013**, *52*, 6464. (b) Schouwey, C.; Holstein, J. J.; Scopelliti, R.; Zhurov, K. O.; Nagornov, K. O.; Tsybin, Y. O.; Smart, O. S.; Bricogne, G.; Severin, K. *Angew. Chem., Int. Ed.* **2014**, *53*, 11261. (c) Chichak, K. S.; Cantrill, S. J.; Pease, A. R.; Chiu, S.-H.; Cave, G. W.; Atwood, J. L.; Stoddart, J. F. *Science* **2004**, *304*, 1308.
- (20) (a) Yoshizawa, M.; Tamura, M.; Fujita, M. *Science* **2006**, *312*, 251. (b) Nishioka, Y.; Yamaguchi, T.; Kawano, M.; Fujita, M. *J. Am. Chem. Soc.* **2008**, *130*, 8160. (c) Howlader, P.; Das, P.; Zangrando, E.; Mukherjee, P. S. *J. Am. Chem. Soc.* **2016**, *138*, 1668. (d) Wang, Q.-Q.; Gonell, S.; Leenders, S. H.; Dürr, M.; Ivanović-Burmazović, I.; Reek, J. N. *Nat. Chem.* **2016**, *8*, 225. (e) Cullen, W.; Misuraca, M. C.; Hunter, C. A.; Williams, N. H.; Ward, M. D. *Nat. Chem.* **2016**, *8*, 231. (f) Marcos, V.; Stephens, A. J.; Jaramillo-Garcia, J.; Nussbaumer, A. L.; Woltering, S. L.; Valero, A.; Lemonnier, J.-F.; Vitorica-Yrezabal, I. J.; Leigh, D. A. *Science* **2016**, *352*, 1555. (g) Wang, Z. J.; Clary, K. N.; Bergman, R. G.; Raymond, K. N.; Toste, F. D. *Nat. Chem.* **2013**, *5*, 100. (h) Pluth, M. D.; Bergman, R. G.; Raymond, K. N. *Science* **2007**, *316*, 85.
- (21) (a) Masar, M. S.; Gianneschi, N. C.; Oliveri, C. G.; Stern, C. L.; Nguyen, S. T.; Mirkin, C. A. *J. Am. Chem. Soc.* **2007**, *129*, 10149. (b) Roy, B.; Ghosh, A. K.; Srivastava, S.; D'Silva, P.; Mukherjee, P. S. *J. Am. Chem. Soc.* **2015**, *137*, 11916. (c) Chen, L.-J.; Ren, Y.-Y.; Wu, N.-W.; Sun, B.; Ma, J.-Q.; Zhang, L.; Tan, H.; Liu, M.; Li, X.; Yang, H.-B. *J. Am. Chem. Soc.* **2015**, *137*, 11725.
- (22) (a) Samanta, S. K.; Moncelet, D.; Briken, V.; Isaacs, L. J. *Am. Chem. Soc.* **2016**, *138*, 14488. (b) Zheng, Y.-R.; Suntharalingam, K.; Johnstone, T. C.; Lippard, S. J. *Chem. Sci.* **2015**, *6*, 1189.
- (23) Schwamborn, K.; Caprioli, R. M. *Nat. Rev. Cancer* **2010**, *10*, 639.
- (24) (a) Li, Z.-Y.; Zhang, Y.; Zhang, C.-W.; Chen, L.-J.; Wang, C.; Tan, H.; Yu, Y.; Li, X.; Yang, H.-B. *J. Am. Chem. Soc.* **2014**, *136*, 8577. (b) Shi, Y.; Wang, M.; Ma, C.; Wang, Y.; Li, X.; Yu, G. *Nano Lett.* **2015**, *15*, 6276.
- (25) (a) Yan, X.; Xu, J.-F.; Cook, T. R.; Huang, F.; Yang, Q.-Z.; Tung, C.-H.; Stang, P. J. *Proc. Natl. Acad. Sci. U. S. A.* **2014**, *111*, 8717. (b) Yan, X.; Cook, T. R.; Pollock, J. B.; Wei, P.; Zhang, Y.; Yu, Y.; Huang, F.; Stang, P. J. *J. Am. Chem. Soc.* **2013**, *135*, 14036.
- (26) (a) Mal, P.; Breiner, B.; Rissanen, K.; Nitschke, J. R. *Science* **2009**, *324*, 1697. (b) Fiedler, D.; Bergman, R. G.; Raymond, K. N. *Angew. Chem., Int. Ed.* **2006**, *45*, 745. (c) Sawada, T.; Yoshizawa, M.; Sato, S.; Fujita, M. *Nat. Chem.* **2009**, *1*, 53.
- (27) (a) Samanta, S. K.; Rana, A.; Schmittel, M. *Angew. Chem., Int. Ed.* **2016**, *55*, 2267. (b) Saha, M. L.; Schmittel, M. *J. Am. Chem. Soc.* **2013**, *135*, 17743. (c) Mahata, K.; Schmittel, M. *J. Am. Chem. Soc.* **2009**, *131*, 16544. (d) Mahata, K.; Saha, M. L.; Schmittel, M. *J. Am. Chem. Soc.* **2010**, *132*, 15933. (e) Ronson, T. K.; Roberts, D. A.; Black, S. P.; Nitschke, J. R. *J. Am. Chem. Soc.* **2015**, *137*, 14502. (f) Smulders, M. M.; Jiménez, A.; Nitschke, J. R. *Angew. Chem., Int. Ed.* **2012**, *51*, 6681. (g) Ramsay, W. J.; Szczypiński, F. T.; Weissman, H.; Ronson, T. K.; Smulders, M. M. J.; Rybtchinski, B.; Nitschke, J. R. *Angew. Chem., Int. Ed.* **2015**, *54*, 5636.
- (28) (a) Sun, Q.-F.; Murase, T.; Sato, S.; Fujita, M. *Angew. Chem., Int. Ed.* **2011**, *50*, 10318. (b) Bhat, I. A.; Samanta, D.; Mukherjee, P. S. *J. Am. Chem. Soc.* **2015**, *137*, 9497. (c) Wang, M.; Wang, C.; Hao, X.-Q.; Liu, J.; Li, X.; Xu, C.; Lopez, A.; Sun, L.; Song, M.-P.; Yang, H.-B.; Li, X. *J. Am. Chem. Soc.* **2014**, *136*, 6664. (d) Wang, S.-Y.; Fu, J.-H.; Liang, Y.-P.; He, Y.-J.; Chen, Y.-S.; Chan, Y.-T. *J. Am. Chem. Soc.* **2016**, *138*, 3651. (e) Fu, J.-H.; Lee, Y. H.; He, Y. J.; Chan, Y. T. *Angew. Chem., Int. Ed.* **2015**, *54*, 6231. (f) Song, B.; Zhang, Z.; Wang, K.; Hsu, C.-H.; Bolarinwa, O.; Wang, J.; Li, Y.; Yin, G.-Q.; Rivera, E.; Yang, H.-B.; Liu, C.; Xu, B.; Li, X. *Angew. Chem., Int. Ed.* **2017**, *56*, 5258. (g) Wang, M.; Wang, K.; Wang, C.; Huang, M.; Hao, X.-Q.; Shen, M.-Z.; Shi, G.-Q.; Zhang, Z.; Song, B.; Cisneros, A.; Song, M.-P.; Xu, B.; Li, X. *J. Am. Chem. Soc.* **2016**, *138*, 9258.
- (29) (a) Sepehrpour, H.; Saha, M. L.; Stang, P. J. *J. Am. Chem. Soc.* **2017**, *139*, 2553. (b) Li, K.; Zhang, L.-Y.; Yan, C.; Wei, S.-C.; Pan, M.; Zhang, L.; Su, C.-Y. *J. Am. Chem. Soc.* **2014**, *136*, 4456. (c) Newkome, G. R.; Wang, P.; Moorefield, C. N.; Cho, T. J.; Mohapatra, P. P.; Li, S.; Hwang, S.-H.; Lukyanova, O.; Echegoyen, L.; Palagallo, J. A.; Iancu, V.; Hla, S.-W. *Science* **2006**, *312*, 1782. (d) Lu, X.; Li, X.; Guo, K.; Xie, T.-Z.; Moorefield, C. N.; Wesdemiotis, C.; Newkome, G. R. *J. Am. Chem. Soc.* **2014**, *136*, 18149. (e) Li, Y.; Jiang, Z.; Wang, M.; Yuan, J.; Liu, D.; Yang, X.; Chen, M.; Yan, J.; Li, X.; Wang, P. *J. Am. Chem. Soc.* **2016**, *138*, 10041. (f) Ramsay, W. J.; Rizzuto, F. J.; Ronson, T. K.; Caprice, K.; Nitschke, J. R. *J. Am. Chem. Soc.* **2016**, *138*, 7264.
- (30) (a) Granzhan, A.; Schouwey, C.; Riis-Johannessen, T.; Scopelliti, R.; Severin, K. *J. Am. Chem. Soc.* **2011**, *133*, 7106. (b) Mal, P.; Schultz, D.; Beyeh, K.; Rissanen, K.; Nitschke, J. R. *Angew. Chem., Int. Ed.* **2008**, *47*, 8297. (c) Meng, W.; Breiner, B.; Rissanen, K.; Thoburn, J. D.; Clegg, J. K.; Nitschke, J. R. *Angew. Chem., Int. Ed.* **2011**, *50*, 3479. (d) Bilbeisi, R. A.; Clegg, J. K.; Elgrishi, N.; Hatten, X. d.; Devillard, M.; Breiner, B.; Mal, P.; Nitschke, J. R. *J. Am. Chem. Soc.* **2012**, *134*, 5110. (e) Bilbeisi, R. A.; Ronson, T. K.; Nitschke, J. R. *Angew. Chem., Int. Ed.* **2013**, *52*, 9027.
- (31) (a) Black, S. P.; Stefankiewicz, A. R.; Smulders, M. M.; Sattler, D.; Schalley, C. A.; Nitschke, J. R.; Sanders, J. K. *Angew. Chem., Int. Ed.* **2013**, *52*, 5749. (b) Li, S.; Huang, J.; Zhou, F.; Cook, T. R.; Yan, X.; Ye, Y.; Zhu, B.; Zheng, B.; Stang, P. J. *J. Am. Chem. Soc.* **2014**, *136*, 5908. (c) Park, K.-M.; Kim, S.-Y.; Heo, J.; Whang, D.; Sakamoto, S.; Yamaguchi, K.; Kim, K. *J. Am. Chem. Soc.* **2002**, *124*, 2140. (d) Zhukhovitskiy, A. V.; Zhong, M.; Keeler, E. G.; Michaelis, V. K.; Sun, J. E.; Hore, M. J.; Pochan, D. J.; Griffin, R. G.; Willard, A. P.; Johnson, J. A. *Nat. Chem.* **2016**, *8*, 33.
- (32) (a) Eryazici, I.; Moorefield, C. N.; Newkome, G. R. *Chem. Rev.* **2008**, *108*, 1834. (b) Constable, E. C. *Chem. Soc. Rev.* **2007**, *36*, 246. (c) Hofmeier, H.; Schubert, U. S. *Chem. Soc. Rev.* **2004**, *33*, 373. (d) Wild, A.; Winter, A.; Schlütter, F.; Schubert, U. S. *Chem. Soc. Rev.* **2011**, *40*, 1459.
- (33) Meier, M. A.; Lohmeijer, B. G.; Schubert, U. S. *J. Mass Spectrom.* **2003**, *38*, 510.
- (34) Newkome, G. R.; Moorefield, C. N. *Chem. Soc. Rev.* **2015**, *44*, 3954.

- (35) (a) Chan, Y. T.; Li, X.; Moorefield, C. N.; Wesdemiotis, C.; Newkome, G. R. *Chem. - Eur. J.* **2011**, *17*, 7750. (b) Schultz, A.; Li, X.; McCusker, C. E.; Moorefield, C. N.; Castellano, F. N.; Wesdemiotis, C.; Newkome, G. R. *Chem. - Eur. J.* **2012**, *18*, 11569. (c) Schultz, A.; Li, X.; Barkakaty, B.; Moorefield, C. N.; Wesdemiotis, C.; Newkome, G. R. *J. Am. Chem. Soc.* **2012**, *134*, 7672. (d) Liu, D.; Jiang, Z.; Wang, M.; Yang, X.; Liu, H.; Chen, M.; Moorefield, C. N.; Newkome, G. R.; Li, X.; Wang, P. *Chem. Commun.* **2016**, *52*, 9773.
- (36) Yang, H.-B.; Hawkridge, A. M.; Huang, S. D.; Das, N.; Bunge, S. D.; Muddiman, D. C.; Stang, P. J. *J. Am. Chem. Soc.* **2007**, *129*, 2120.
- (37) Wang, J.-L.; Li, X.; Lu, X.; Chan, Y.-T.; Moorefield, C. N.; Wesdemiotis, C.; Newkome, G. R. *Chem. - Eur. J.* **2011**, *17*, 4830.
- (38) (a) Fermi, A.; Bergamini, G.; Roy, M.; Gingras, M.; Ceroni, P. *J. Am. Chem. Soc.* **2014**, *136*, 6395. (b) Bauer, T.; Zheng, Z.; Renn, A.; Enning, R.; Stemmer, A.; Sakamoto, J.; Schlüter, A. D. *Angew. Chem., Int. Ed.* **2011**, *50*, 7879. (c) Zheng, Z.; Opilik, L.; Schiffmann, F.; Liu, W.; Bergamini, G.; Ceroni, P.; Lee, L.-T.; Schütz, A.; Sakamoto, J.; Zenobi, R.; Van de Vondelle, J.; Schlüter, A. D. *J. Am. Chem. Soc.* **2014**, *136*, 6103. (d) Takada, K.; Sakamoto, R.; Yi, S.-T.; Katagiri, S.; Kambe, T.; Nishihara, H. *J. Am. Chem. Soc.* **2015**, *137*, 4681.
- (39) (a) Bauer, T.; Schlüter, A. D.; Sakamoto, J. *Synlett* **2010**, *2010*, 877. (b) Wang, J.-L.; Li, X.; Lu, X.; Hsieh, I.-F.; Cao, Y.; Moorefield, C. N.; Wesdemiotis, C.; Cheng, S. Z.; Newkome, G. R. *J. Am. Chem. Soc.* **2011**, *133*, 11450.
- (40) (a) Wang, J.-L.; Li, X.; Lu, X.; Hsieh, I. F.; Cao, Y.; Moorefield, C. N.; Wesdemiotis, C.; Cheng, S. Z. D.; Newkome, G. R. *J. Am. Chem. Soc.* **2011**, *133*, 11450. (b) Lu, X.; Li, X.; Cao, Y.; Schultz, A.; Wang, J.-L.; Moorefield, C. N.; Wesdemiotis, C.; Cheng, S. Z. D.; Newkome, G. R. *Angew. Chem., Int. Ed.* **2013**, *52*, 7728.
- (41) (a) Giuseppone, N.; Schmitt, J.-L.; Allouche, L.; Lehn, J.-M. *Angew. Chem., Int. Ed.* **2008**, *47*, 2235. (b) Schulze, B. M.; Watkins, D. L.; Zhang, J.; Ghiviriga, I.; Castellano, R. K. *Org. Biomol. Chem.* **2014**, *12*, 7932.
- (42) (a) Bocker, E. R.; Anderson, S. E.; Northrop, B. H.; Stang, P. J.; Bowers, M. T. *J. Am. Chem. Soc.* **2010**, *132*, 13486. (b) Chan, Y.-T.; Li, X.; Soler, M.; Wang, J.-L.; Wesdemiotis, C.; Newkome, G. R. *J. Am. Chem. Soc.* **2009**, *131*, 16395. (c) Thiel, J.; Yang, D.; Rosnes, M. H.; Liu, X.; Yvon, C.; Kelly, S. E.; Song, Y. F.; Long, D. L.; Cronin, L. *Angew. Chem., Int. Ed.* **2011**, *50*, 8871. (d) Ujma, J.; De Cecco, M.; Chepelin, O.; Levene, H.; Moffat, C.; Pike, S. J.; Lusby, P. J.; Barran, P. E. *Chem. Commun.* **2012**, *48*, 4423.
- (43) (a) Li, X.; Chan, Y.-T.; Newkome, G. R.; Wesdemiotis, C. *Anal. Chem.* **2011**, *83*, 1284. (b) Perera, S.; Li, X.; Soler, M.; Schultz, A.; Wesdemiotis, C.; Moorefield, C. N.; Newkome, G. R. *Angew. Chem., Int. Ed.* **2010**, *49*, 6539.
- (44) (a) Shvartsburg, A. A.; Jarrold, M. F. *Chem. Phys. Lett.* **1996**, *261*, 86. (b) Jarrold, M. F. *Annu. Rev. Phys. Chem.* **2000**, *51*, 179. (c) Shvartsburg, A. A.; Liu, B.; Siu, K. M.; Ho, K.-M. *J. Phys. Chem. A* **2000**, *104*, 6152.
- (45) (a) Sun, B.; Wang, M.; Lou, Z.; Huang, M.; Xu, C.; Li, X.; Chen, L.-J.; Yu, Y.; Davis, G. L.; Xu, B.; Yang, H.-B.; Li, X. *J. Am. Chem. Soc.* **2015**, *137*, 1556. (b) Wang, M.; Wang, K.; Wang, C.; Huang, M.; Hao, X.-Q.; Shen, M.-Z.; Shi, G.-Q.; Zhang, Z.; Song, B.; Cisneros, A.; Song, M.-P.; Xu, B.; Li, X. *J. Am. Chem. Soc.* **2016**, *138*, 9258.
- (46) (a) Shukla, R. B. *J. Magn. Reson., Ser. A* **1995**, *113*, 196. (b) Shokhireva, T. K.; Nasset, M. J. M.; Ann Walker, F. *Inorg. Chim. Acta* **1998**, *272*, 204.
- (47) Zheng, J.-Y.; Tashiro, K.; Hirabayashi, Y.; Kinbara, K.; Saigo, K.; Aida, T.; Sakamoto, S.; Yamaguchi, K. *Angew. Chem., Int. Ed.* **2001**, *40*, 1857.
- (48) (a) Nazeeruddin, M. K.; Péchy, P.; Renouard, T.; Zakeeruddin, S. M.; Humphry-Baker, R.; Comte, P.; Liska, P.; Cevey, L.; Costa, E.; Shklover, V.; Spiccia, L.; Deacon, G. B.; Bignozzi, C. A.; Grätzel, M. *J. Am. Chem. Soc.* **2001**, *123*, 1613. (b) Satoh, N.; Cho, J.-S.; Higuchi, M.; Yamamoto, K. *J. Am. Chem. Soc.* **2003**, *125*, 8104. (c) Wang, Q.; Zakeeruddin, S. M.; Cremer, J.; Bäuerle, P.; Humphry-Baker, R.; Grätzel, M. *J. Am. Chem. Soc.* **2005**, *127*, 5706. (d) Liu, C.-L.; Zheng, C.-J.; Liu, X.-K.; Chen, Z.; Yang, J.-P.; Li, F.; Ou, X.-M.; Zhang, X.-H. *J. Mater. Chem. C* **2015**, *3*, 1068. (e) Watanabe, Y.; Yoshioka, R.; Sasabe, H.; Kamata, T.; Katagiri, H.; Yokoyama, D.; Kido, J. *J. Mater. Chem. C* **2016**, *4*, 8980.
- (49) Robson, K. C. D.; Koivisto, B. D.; Gordon, T. J.; Baumgartner, T.; Berlinguette, C. P. *Inorg. Chem.* **2010**, *49*, 5335.